

ISLAMIC UNIVERSITY OF TECHNOLOGY



OPERATING SYSTEM LAB
CSE 4502

Lab 1 Assignment

Submitted by:

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Section: 1B

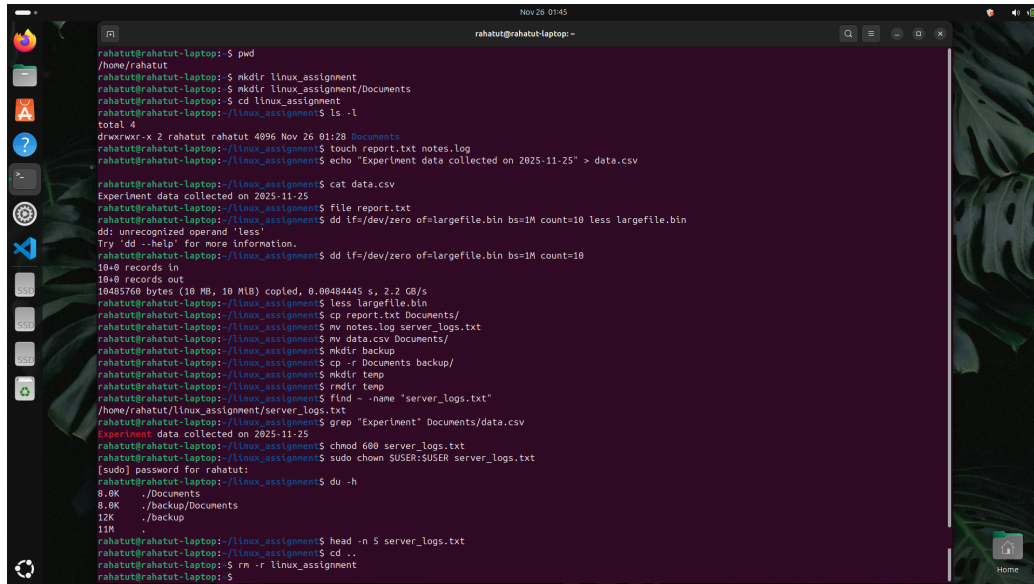
1. List of Commands

1. **pwd** — Print Working Directory (verify your current location)
2. **mkdir** — Make directories (e.g., `linux_assignment`, `Documents`)
3. **cd** — Change directory
4. **ls** — List directory contents (`ls -l` for details)
5. **touch** — Create empty files or update timestamps
6. **echo** — Output text (e.g., redirect to `data.csv`)
7. **cat** — Display file content
8. **file** — Show file type
9. **less** — View files page by page
10. **cp** — Copy files
11. **mv** — Move or rename files/directories
12. **cp -r** — Copy directories recursively
13. **rmdir** — Remove empty directories
14. **find** — Search for files in a directory hierarchy
15. **grep** — Search for text patterns in files
16. **chmod** — Change file permissions
17. **chown** — Change file owner/group
18. **du** — Check disk usage of files/directories
19. **head** — Show the first few lines of a file
20. **rm** — Remove files or directories

2. Exact Command Syntax

```
1 pwd
2 mkdir linux_assignment
3 mkdir linux_assignment/Documents
4 cd linux_assignment
5 ls -l
6 touch report.txt notes.log
7 echo "Experiment data collected on 2025-11-25" > data.csv
8 cat data.csv
9 file report.txt
10 dd if=/dev/zero of=largefile.bin bs=1M count=10
11 less largefile.bin
12 cp report.txt Documents/
13 mv notes.log server_logs.txt
14 mv data.csv Documents/
15 mkdir backup
16 cp -r Documents backup/
17 mkdir temp
18 rmdir temp
19 find ~ -name "server_logs.txt"
20 grep "Experiment" Documents/data.csv
21 chmod 600 server_logs.txt
22 sudo chown $USER:$USER server_logs.txt
23 du -h
24 head -n 5 server_logs.txt
25 cd ..
26 rm -r linux_assignment
```

3. Execution Proof



```
rahatur@rahatur-laptop:~$ pwd
/home/rahatur
rahatur@rahatur-laptop:~$ mkdir linux_assignment
rahatur@rahatur-laptop:~$ mkdir linux_assignment/Documents
rahatur@rahatur-laptop:~$ cd linux_assignment
rahatur@rahatur-laptop:~/linux_assignment$ ls -l
total 4
drwxrwxr-x 2 rahatur rahatur 4096 Nov 26 01:28 Documents
rahatur@rahatur-laptop:~/linux_assignment$ touch report.txt notes.log
rahatur@rahatur-laptop:~/linux_assignment$ echo "Experiment data collected on 2025-11-25" > data.csv
rahatur@rahatur-laptop:~/linux_assignment$ cat data.csv
Experiment data collected on 2025-11-25
rahatur@rahatur-laptop:~/linux_assignment$ file report.txt
report.txt: ASCII text
rahatur@rahatur-laptop:~/linux_assignment$ dd if=/dev/zero of=largefile.bin bs=1M count=10 less largefile.bin
dd: unrecognized operand 'less'
Try 'dd --help' for more information.
rahatur@rahatur-laptop:~/linux_assignment$ dd if=/dev/zero of=largefile.bin bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.00484445 s, 2.2 GB/s
rahatur@rahatur-laptop:~/linux_assignment$ less largefile.bin
rahatur@rahatur-laptop:~/linux_assignment$ cp report.txt Documents/
rahatur@rahatur-laptop:~/linux_assignment$ mv notes.log server_logs.txt
rahatur@rahatur-laptop:~/linux_assignment$ mv data.csv Documents/
rahatur@rahatur-laptop:~/linux_assignment$ mkdir backup
rahatur@rahatur-laptop:~/linux_assignment$ cp -r Documents backup/
rahatur@rahatur-laptop:~/linux_assignment$ mkdir temp
rahatur@rahatur-laptop:~/linux_assignment$ rmdir temp
rahatur@rahatur-laptop:~/linux_assignment$ find -name "server_logs.txt"
/home/rahatur/linux_assignment/server_logs.txt
rahatur@rahatur-laptop:~/linux_assignment$ grep "Experiment" Documents/data.csv
Experiment data collected on 2025-11-25
rahatur@rahatur-laptop:~/linux_assignment$ chmod 600 server_logs.txt
rahatur@rahatur-laptop:~/linux_assignment$ sudo chown $USER:$USER server_logs.txt
[sudo] password for rahatur:
rahatur@rahatur-laptop:~/linux_assignment$ du -h
8.0K  ./Documents
8.0K  ./backup/Documents
12K   ./backup
11M
rahatur@rahatur-laptop:~/linux_assignment$ head -n 5 server_logs.txt
rahatur@rahatur-laptop:~/linux_assignment$ cd ..
rahatur@rahatur-laptop:~$ rm -r linux_assignment
rahatur@rahatur-laptop:~$
```

Figure 1: Terminal execution of all 20 commands.